

11.5 NORMATIVE MEASURES

The measures that essentially involve judgement about values through specification of social welfare function are called normative measures. The arguments of this nature were first advanced by Dalton, pretty eight decades ago in 1920 for constructing what are today called normative measures of inequality.

Reacting to an observation by Pearsons (1909) that 'the statistical problem before the economists in determining upon a measure of inequality in the distribution of wealth is identical with that of the biologist in determining upon a measure of the inequality in the distribution of any physical characteristic', Dalton (1920) pointed out that 'economist is interested, not in distribution as such, but in effects of the distribution upon the distribution (and total amount) of economic welfare which may be derived from income'. The objection to great inequality of income, he further points out, is due to the resulting loss of potential economic welfare that could accrue to people in the absence of it.

Yet, it has to be noted that inequality though defined in terms of economic welfare, has to be measured in terms of income. This idea due to Dalton has been conceded by subsequent contributions. Using the notion of social welfare function in construction gives rise to **normative** measures of inequality.

It may be instructive to remember that the discussion would revolve around three issues:

- 1) the relationship between income of a person and his welfare;
- 2) the relationship between personal income-welfare functions; and
- 3) the relationship between personal welfare and social welfare.

It may be noted that utility is the word mostly used for personal welfare whereas for welfare of society the phrase social utility is rarely used.

There are two major indices in this category: Dalton's index and Atkinson's index. In Atkinson's index a new idea is introduced and that is of equally distributed equivalent income. Actually there are two sub-approaches within normative approach. One is Dalton's and the other is Atkinson's. While in Dalton's approach present social welfare is compared with that could be obtained by equally distributing the total income, in Atkinson's approach the present level of income is compared with that of equally distributed level of income, which generates the present level of social welfare. Sen has generalised the Atkinson's index. Theil's index based on information theory could be suggested here only to sort of complete the unit.

11.5.1 Dalton Index

For each individual, Dalton assumes, marginal economic welfare diminishes as income increases. It means income-welfare function

$$U_i = U_i(x_i), i = 1, 2, \dots, N \quad (\text{D.1})$$

(where U_i is welfare of person i possessing income x_i)

is concave, suggesting that $(\partial U_i / \partial x_i) > 0$ but $(\partial^2 U_i / \partial x_i^2) < 0$. Dalton further assumes that economic welfare of different persons is additive. Thus, in his scheme, social welfare is a simple aggregation of personal welfares. In other words, social welfare W is given by

$$W = \sum_{i=1}^N U_i(x_i) \quad (D.2)$$

He further assumes that the relation of income to economic welfare is the same for all members of the community. That is,

$$U_i = U(x_i), i=1,2,\dots,N \quad (D.3)$$

In that case, the relation (D.2) can be expressed as

$$W = \sum_{i=1}^N U(x_i) \quad (D.4)$$

which makes it clear that whosoever gains in welfare, the addition to the social welfare is the same. For any given level of social welfare, any distribution of welfare among the members of the society is permissible. However, one must remember that the relation of individual income to their welfares is concave. Therefore, transfer of income from A to B will not lead to symmetric change in welfares of those two persons involved in the transaction. The result is some impact on W the measure of social welfare.

From Fig. 11.7, we may compare the situation when two individuals, both possessing the same relation, have two different income levels, with that when they have the same (mean) income. We may note that the sum of the welfare of person 1 (BB') and the welfare of person 2 (DD') is less than the twice of CC' which is the level of welfare enjoyed by both the persons when they have equal income. It is easy to see that the loss suffered by person 2, that is D'E, is overcompensated by the gained by person 1, which is C'F.

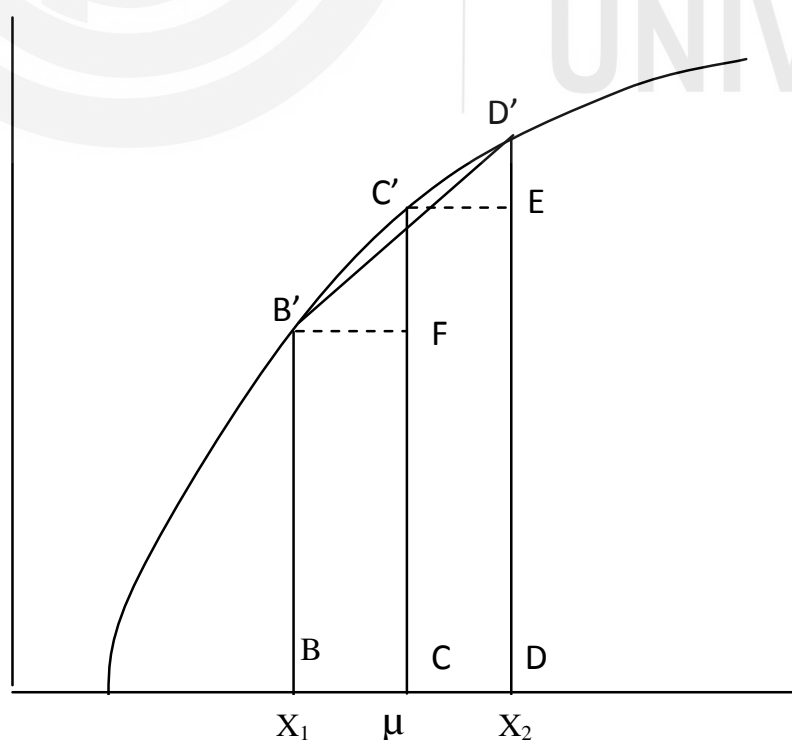


Fig. 11.7

This demonstrates that, under assumptions by Dalton, an equal distribution is preferable to an unequal one for a given amount of total income from the viewpoint of social welfare. In fact, for a given total of income, the economic welfare of the society will be maximum when all incomes are equal. The inequality of any given distribution may therefore be defined as

$$D_1 = \frac{\sum_{i=1}^N U(\mu)}{\sum_{i=1}^N U(x_i)} = \frac{NU(\mu)}{\sum_{i=1}^N U(x_i)} \quad (D.7)$$

which is equal to unity for an equal distribution and greater than unity for an unequal one. It may therefore be preferred to define the Dalton's index as

$$D_2 = \frac{NU(\mu)}{\sum_{i=1}^N U(x_i)} \quad (D.8)$$

which is obviously zero for an equal distribution. How large can it be? It will depend on the values of $U(0)$, $U(\mu)$ and $U(N\mu)$ when N and μ are given, not necessarily 1. Later writers have therefore preferred to define Dalton's index in the following form, which inverts the arguments of D_2 subtract it from 1:

$$D = 1 - \frac{\sum_{i=1}^N U(x_i)}{NU(\mu)} = 1 - \frac{\bar{U}}{U(\mu)} \quad (D.9)$$

It looks as if the index is contained in the interval (0,1). However, there are many valid concave functions where it may not hold true. For example, if we have $U(x_i) = \log x_i$, then $D = 1 - \{\log \hat{\mu} / \log \mu\}$. Given the fact that $\hat{\mu} < \mu$, D would turn out to be a negative number for $\mu < 1$. And μ could be less than 1 as x can be measured in any unit. It would be the same case $U(x_i) = 1/x_i$.

However, in order to obtain numerical magnitude, it is not sufficient to define the index. Dalton (1920) points out that though defined in terms of economic welfare, inequality has to be measured in terms of income. Then, no unique measure of inequality will emerge. It will verily depend on the particular functional relationship assumed. Dalton himself considered two such functions for the purpose of illustration. The first is related to Bernaulli's hypothesis. It holds that proportionate additions to income (in excess of that required for bare subsistence-poverty line) make equal additions to personal welfare, That is,

$$dU_i = \frac{dx_i}{x_i} \text{ or } U_i = \log x_i + c_i \quad (D.10)$$

Under the assumption that every person has the same functional relationship, the Dalton's index can be given as

$$D = 1 - \frac{\log \hat{\mu} + c}{\log \mu + c} \quad (D.11)$$

where $\hat{\mu}$ is the geometric mean of personal incomes. The other formulation he discusses is given as

$$dU_i = \frac{dx_i}{x_i^2} \text{ or } U_i = c - \frac{1}{x_i} \quad (\text{D.12})$$

where c is the maximum welfare one can obtain when $x \rightarrow \infty$. Dalton's index in this case would turn out to be:

$$D = 1 - \frac{c - (1/\tilde{\mu})}{C - (1/\mu)} \quad (\text{D.13})$$

where $\tilde{\mu}$ is the harmonic mean.

11.5.2 Atkinson Index

Atkinson (1970) objects to Dalton's measure because D is not invariant with respect to positive linear transformations of personal income-welfare functions. This was pointed out by Dalton himself but he could not resolve it.

Atkinson seeks to redefine the index in such a way that measurement would be invariant with respect to permitted transformations of welfare numbers. Atkinson does it through devising what he calls 'equally distributed equivalent income'. Both the distributions, the original and the new one, are supposed to yield the same level of welfare.

In order to make the concept clear, we put a few artifacts along with the actual distribution. Let us first note that for an actually distributed income vector x_i , $i=1,2,\dots,N$ (call it vector a), there is only one equally distributed income vector with each element equal to μ (call it vector b) but there are a number of equivalently distributed, vectors (call them vectors c). See Chart 1. An equivalent income distribution is one, which has the same level of welfare as that of currently given distribution. However, one of these equivalent distributions (vectors c) is 'equal' as well. This is called equally distributed equivalent income vector, shown as vector (d) in the Chart 1. As μ is the mean level of current distribution, μ^* may be used for designating the level of equally distributed equivalent income.

CHART-I

Vector (a) Actually distributed income vector	$x_1, x_2, \dots, x_i, \dots, x_N.$
Vector (b) Equally distributed income vector	$\mu, \mu, \dots, \mu, \dots, \mu.$
Vector (c) Equivalently distributed income vector	$x_1^*, x_2^*, \dots, x_i^*, \dots, x_N^*.$
Vector (d) Equally distributed equivalent income vector	$\mu^*, \mu^*, \dots, \mu^*, \dots, \mu^*$

It should be obvious that $W_b \geq W_a = W_c$ and $W_c = W_d$. Then, $W_a = W_d$. W represents social welfare with respective distributions of income vectors. It is clear that $\mu \geq \mu^*$. μ^* is defined by the additive social welfare function having symmetric individual utility functions such as:

$$U(\mu) = \frac{1}{N} \sum_{i=1}^N U(x_i) \quad (\text{A.1})$$

or equivalently

$$\mu^* = U^{-1} \left[\frac{1}{N} \sum_{i=1}^N U(x_i) \right] \quad (\text{A.2})$$

The index due to Atkinson is then defined as the additive inverse of the ratio of equivalent mean income to actual mean income:

$$A = 1 - \frac{\mu^*}{\mu} \quad (\text{A.3})$$

which is said to lie between zero (complete equality) and 1 (complete inequality). We can see that A cannot be 1 unless μ^* is zero, which is an impossibility for any distribution with $\mu > 0$. If complete inequality is defined as the situation when only one person grabs all the income, we can see that

$$A = 1 - \frac{\mu_m^*}{\mu} \quad (\text{A.4})$$

where

$$NU(\mu^*) = \sum_{i=1}^N U(x_i)$$

and

$$NU(\mu_m^*) = (N - 1)U(0) + U(N\mu).$$

This index is not scale independent unless some restriction is imposed on the relationship U . If this requirement has to be met, Atkinson points out, we may have to have the following form

$$U(x_i) = \begin{cases} \alpha + \frac{\beta}{1-\epsilon} x_i^{1-\epsilon}, & \epsilon \neq 1 \\ \log_e x_i, & \epsilon = 1 \end{cases} \quad (\text{A.5})$$

Note that we need $\epsilon \geq 0$ for ensuring concavity and $\epsilon > 0$ for ensuring strict concavity. This is a homothetic function and is linear when $\epsilon = 0$. We may note that ϵ cannot exceed 1 as in that case the varying component assumes inverse relationship. α is usually negative so that $U(x_i)$ is not positive for $x_i = 0$. Otherwise, when $x_i = 0$, $U_i = \alpha$ which means that welfare is positive even when income is zero. This is generally not acceptable. On the contrary, a negative α would be more acceptable. When $\epsilon = 1$, α is infinitely large and negative.

Since ϵ can be zero, Atkinson's requirement is not strict concavity. Sen (1973) has a question. He asks to consider two distributions (0,10) and (5,5) along with

$$U(x_i) = \alpha + \beta(x_i) \quad (\text{A.6})$$

Then, he points out the level of social welfare would be $(2\alpha + 10\beta)$ whatever the distribution. μ^* would be 5 in both the cases. μ is of course 5. The measure of inequality A is therefore zero. So, both the distributions are ethically equal. This is obviously absurd. Therefore, the relation (A.5) should be defined with the restriction $\epsilon > 0$. We should also note that (A.4) is an iso-elastic marginal utility function.

ϵ is the inequality-aversion parameter and has close resemblance with risk-aversion premium. Atkinson proposed to draw on the parallel formally with the problem of measuring risk. He finds his concept of equally distributed equivalent income very closely resembles with risk-premium or certainty equivalent income as used in the theory of decision-making under uncertainty.

In case we seek to introduce this restrictive personal income-welfare function along with simple aggregation of individual welfares to constitute the social welfare, into the inequality measure A , we will have

$$A = 1 - \left[\frac{1}{N} \sum_{i=1}^N \left(\frac{x_i}{\mu} \right)^{1-\epsilon} \right]^{1/(1-\epsilon)}, \epsilon \neq 1 \quad (\text{A.7})$$

The question is now narrowed down to choosing ϵ . As ϵ rises, more weight is attached to transfers at the lower end of the distribution and less to that at the top. When ϵ rises, (A.7) assumes the function $\min_i (x_i)$, which only takes account of transfers to the very lowest income group (and is therefore not strictly concave). When $\epsilon = 0$, U_i is linear. As a consequence, A is always zero. This means A has no descriptive content at all. When $\epsilon \rightarrow 1$, A turns out to be

$$A = 1 - \prod_{i=1}^N \left(\frac{x_i}{\mu} \right)^{1/N} = 1 - \frac{\hat{\mu}}{\mu} \quad (\text{A.8})$$

which is the same as Champernowne index (CII.1). For values of ϵ between 0 and 1, the expressions may not be very neat. Parameter ϵ is often chosen to be $1/2$ or $1/3$ or $2/3$.

11.5.3 Sen Index

There are people who feel rather strongly that the social valuation of the welfare of individuals should depend crucially on the incomes of their neighbours too. Then, why should society add simply individual welfares? One may also question the assumption of one welfare function for all individuals. If we do so, we should go for broad social welfare function such as

$$W = W(x_1, x_2, \dots, x_N) \quad (\text{S.1})$$

which is just symmetric, quasi-concave and increasing in individual income levels. Then, a more general normative measure of inequality can be defined by devising the concept of 'generalized equally distributed equivalent income'. This is obviously the level of per capita income x^* which, if shared by all, would produce the same level of W as is generated by the present actual distribution. That is,

$$x^* = x | W(x^*, x^*, \dots, x^*) = W(x_1, x_2, \dots, x_N) \quad (\text{S.2})$$

Under the assumption that (S.1) is quasi-concave, $x^* \leq \mu$ for every distribution of income. The index S would then be

$$S = 1 - \frac{x^*}{\mu} \quad (\text{S.3})$$

which is but a generalized version of A. If utilitarian framework is employed, then S and A turn out to be indistinguishable.

These measures, it may be noted, clearly suggest that there exists a redistribution equivalent of growth so far as the concern is about raising the welfare.

11.5.4 Theil Entropy Index

Theil (1967) poses a question: Does information theory supply us with a 'natural' measure of income inequality among N individuals, which is based on income shares? He answers: Yes. Here is a short introduction.

Let us start with income share of individual i :

$$q_i = \frac{x_i}{N_\mu} > 0 \quad \text{such that} \quad \sum_{i=1}^N q_i = 1 \quad (\text{T.1})$$

When $x_i = \mu$, $i=1, 2, \dots, N$, that is, when distribution is equal, we have

$$q_i = \frac{1}{N} \quad i = 1, 2, \dots, N \quad (\text{T.2})$$

We have complete inequality when some $x_i = N\mu$ and $x_j = 0$, $j \neq i$. It implies that $q_i = 1$ for some i and $q_j = 0$, $i \neq j$.

In information theory, one way of defining entropy of probabilities p_i is

$$H = \sum_{i=1}^N p_i \log \frac{1}{p_i}. \quad (\text{T.2})$$

Replacing probabilities by shares, we have

$$H = \sum_{i=1}^N q_i \log \frac{1}{q_i}. \quad (\text{T.3})$$

which can be taken as a measure of equality. For the situation of complete equality, we can see that H is equal to $\log N$ and for that of complete inequality H is zero. We can therefore define Theil index T as

$$\begin{aligned} T &= \log N \sum_{i=1}^N q_i \log \frac{1}{q_i} \\ &= \sum_{i=1}^N q_i \log N - \sum_{i=1}^N q_i \log \frac{1}{q_i} \\ &= \sum_{i=1}^N q_i \log N \cdot q_i \end{aligned} \quad (\text{T.4})$$

This measure is motivated by the notion of entropy in information theory. But one can see that it can be interpreted in the traditional normative framework with

$$U_i = q_i \log \frac{1}{q_i} \quad (\text{T.5})$$

and

$$W = \sum_{i=1}^N U_i(q_i). \quad (\text{T.6})$$

We may note that (T.5) depends on x_i as well as on μ along with N and U that it is concave with respect to x_i .

While the lower limit of T is zero, its upper limit $\log N$ increases as the number of individuals increases. To many people, it is objectionable. However Theil (1967) chooses to defend it. When society consists of two crore persons and one grabs all and when society consists of two persons and one grabs all, cannot have the same level of inequality. The former case is equivalent to the situation in which one crore out of two crore people have nothing and the other one crore have equal income. Maximum value for two-person society is $\log 2$, and that for two crore-person society is $7 \log 2$. Some researchers still insist that the measure should be normalized by dividing it by $\log N$.

Check Your Progress 4

- 1) What is social welfare function, according to Dalton?

.....

.....

.....

.....

.....

- 2) Discuss Dalton index of inequality.

.....

.....

.....

.....

.....

- 3) Give the logic behind Atkinson index.

.....

.....

.....

.....

.....

4) How is Sen index distinct from Atkinson's index?

.....

.....

.....

.....

.....

5) Discuss Theil's entropy index of inequality.

.....

.....

.....

.....

.....

11.6 LET US SUM UP

Owing to adverse impacts of economic inequality both on poverty and on growth, reducing inequality has been a priority of public policy. Various measures of income inequality can be put under two categories: positive measures and normative measures. The positive measures capture the inequality of income without value judgements. These include range quartile range, standard deviation, Gini ratio, etc. Lorenz curve belongs to this category. It measures inequality to the extent of comparing two distributions. The measures, which essentially involve value judgements about social welfare, are normative measures. These include indices propounded by Dalton, Atkinson, Sen, and Theil.

Easy comprehension and easy computation, range of variation and amount of information needed the desirable properties of the measures of economic inequality. In order to judge the efficacy of an inequality index, several axiom have been set up. However, these axioms have been relegated to the Appendix.

11.7 KEY WORDS

- | | | |
|--|---|--|
| Co-efficient of Mean Difference | : | Mean of all pair-wise differences divided by the mean of differences has been termed as coefficient of mean difference in this text. |
| Dispersion | : | The fact that values of a variable are not all the same is known as dispersion. The spread or scattering of the distribution is measured by a measure of dispersion. |

- Extreme Disparity Ratio** : The ratio of the highest value to the lowest value is known as extreme disparity ratio.
- Normative Measures of Inequality** : Measures of inequality, which are articulated through the explicit incorporation of social welfare function or social welfare considerations, are known as the normative measures of inequality.
- Positive Measures of Inequality** : Measures of inequality, which are based in statistical properties of distribution, are known as the positive measures of inequality.
- Relative Standard Deviation** : Standard deviation of a distribution divided by its mean is known as Relative Standard Deviation.
- Standard Logarithmic Deviation** : Standard deviation of logarithms of values in a distribution is known as Standard Logarithmic Deviation. Though logically the deviations of logarithmic values should be taken from the logarithm of geometric mean but at times they are taken from logarithm of arithmetic mean. Therefore, there are two versions.
- Social Welfare Function** : An index of social well-being, often articulated as a function of individual utilities or individual incomes or individual consumption baskets, with or without labour disposition, or individual rankings of potential state of affairs.

11.8 EXERCISES

- 1) Following the adapted distribution of monthly per capita expenditure (in Rs.) in rural India in the 60th round of the NSS over January to June 2004:

Class	50-225	225-255	255-300	300-340	340-380	380-420	420-470	470-525	525-615	615-755	755-950	950-1200
Avg. Exp.	100	240	280	325	365	405	450	500	580	700	850	1100
Percentage of persons	2.4	2.7	6.4	8.3	9.6	9.6	10.8	10.0	12.2	12.6	6.7	8.8

Calculate as many positive measures as you can.

- 2) Following is distribution data of operational holdings from agriculture census 1976-77:

	Holding	Marginal	Small	Semi-medium	Medium	Large	All
Definition	Unit	0.0-1.0 Ha	1.0-2.0 Ha	2.0-4.0 Ha	4.0-10.0 Ha	Above 10.0 Ha	
Number	'000	44523	14728	11666	8212	2440	81569
Area	'000 Ha	17509	20905	32428	49628	42673	163343

Draw Lorenz curve and compute Gini ratio.